Machine Learning — Lecture 4

Logistic Regression and Classification

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Lecture 4 at a Glance

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Logistic Regression and Classification

Course Learning Outcome

CLO3: Construct and analyze machine learning models by applying algorithmic principles and internal computations.

CLO4: Apply supervised learning techniques to solve practical problems and interpret their outcomes.

Lecture purpose

- Move from regression to classification.
- Learn how logistic regression converts a linear score into a probability.
- Understand binary cross-entropy as the classification loss.
- Compute a forward pass, gradients, and one parameter update.
- Interpret thresholds, coefficients, probabilities, and decision boundaries.

Conceptual structure

Phase	Student question
1. Understand	How is a binary probability model built?
2. Train	How do log loss and gradients improve the model?
3. Use & interpret	How should probabilities guide decisions?
4. Extend & apply	How does the model handle new features and tasks?

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💡 Main idea: logistic regression is a classification model that combines a **linear score**, a **sigmoid probability**, and a **loss function suited to binary labels**.

Learning Objectives

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By the end of Lecture 4, you will be able to:

1. Formulate **binary classification** as a supervised-learning problem.
2. Explain why ordinary linear regression is not ideal for classification.
3. Compute a logistic-regression score $z = w^\top x + b$.
4. Convert a score into a probability using the sigmoid function.
5. Apply a decision threshold to obtain a class prediction.
6. Interpret a linear decision boundary and explain how it shifts when the threshold changes.
7. Compute binary cross-entropy loss for one example.
8. Perform one manual gradient-descent update.
9. Explain threshold trade-offs and probability interpretation.
10. Recognize extensions to multiclass classification.

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Lecture Roadmap

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From class probabilities to reliable decisions

Learning phase	Parts	Central question
1. Understand the logistic model	**I–III**	What is binary classification, how do scores become probabilities, and where is the decision boundary?
2. Train the model	**IV–V**	How do binary cross-entropy and gradients determine the parameters?
3. Use and interpret predictions	**VI–VIII**	How do thresholds, coefficient interpretation, and preprocessing affect decisions?
4. Extend and apply	**IX–X**	How do we handle multiclass outcomes and reason through a real classification case?

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💡 **Practice & consolidation:** compute probabilities and losses, compare thresholds, and identify common modeling mistakes.

Part I

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Classification as Supervised Learning

Classification predicts a **category** or **class label**, rather than a continuous numerical value.

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What is a Classification Problem?

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Predicting a class

In supervised **classification**, each training example has: (x_i, y_i),

where:

- x_i is a feature vector;
- y_i is a class label.

For binary classification: $y_i \in \{0, 1\}$

Examples

Problem	Features	Target class
Email filtering	words, sender, links	spam / not spam
Medical screening	age, symptoms, tests	disease / no disease
Fraud detection	amount, merchant, time	fraud / legitimate
Student success	attendance, grades, activity	pass / fail

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💡 The class label is discrete, even if the model internally computes probabilities.

Binary Classification

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Positive and negative classes

Binary classification has two possible outcomes.

We usually choose one class as the **positive class**: $y = 1$

The other class is the **negative class**: $y = 0$

Example — Fraud detection

Class	Label
Fraud	1
Legitimate	0

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Important

The choice of positive class affects interpretation of precision, recall, false positives, and false negatives.

> If fraud is the positive class, a false negative means a fraudulent transaction was missed.

Classification Output

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Class label or probability?

A classifier may output:

1. a class label;
2. a probability estimate;
3. a score that must be converted into a probability or decision.

Logistic regression estimates:

$$P(y = 1 \mid x)$$

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Then a threshold converts probability into a label.

For threshold $t = 0.5$:

$$\hat{y} = \begin{cases} 1, & P(y = 1 \mid x) \geq 0.5 \\ 0, & P(y = 1 \mid x) < 0.5 \end{cases}$$

💡 The model predicts probability first; the threshold turns probability into an action.

Why Not Linear Regression?

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Linear outputs are not probabilities

A **linear regression** model has output:

$$\hat{y} = w^\top x + b$$

For classification, this can be problematic:

- predictions can be less than 0;
- predictions can be greater than 1;
- squared error is not ideal for binary probabilities;
- outliers can distort the line;
- the model does not naturally represent probability.

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Need

We want a model whose output lies in:

$$0 \leq P(y = 1 \mid x) \leq 1$$

Logistic regression solves this by passing a linear score through the sigmoid function.

Part II

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The Logistic Model

Logistic regression applies a sigmoid transformation to a linear score.

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Linear Score

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The same first step as linear models

Logistic regression begins with a **linear score**:

$$z = w^\top x + b$$

For p features:

$$z = w_1 x_1 + w_2 x_2 + \cdots + w_p x_p + b$$

This score can be any real number:

$$z \in (-\infty, +\infty)$$

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Interpretation

- large positive z suggests class 1;
- large negative z suggests class 0;
- $z = 0$ is the **decision boundary** when threshold is 0.5.

Sigmoid Function

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Convert score to probability

The sigmoid function is:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Logistic regression predicts:

$$p = \hat{P}(y = 1 \mid x) = \sigma(w^\top x + b)$$

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Key values

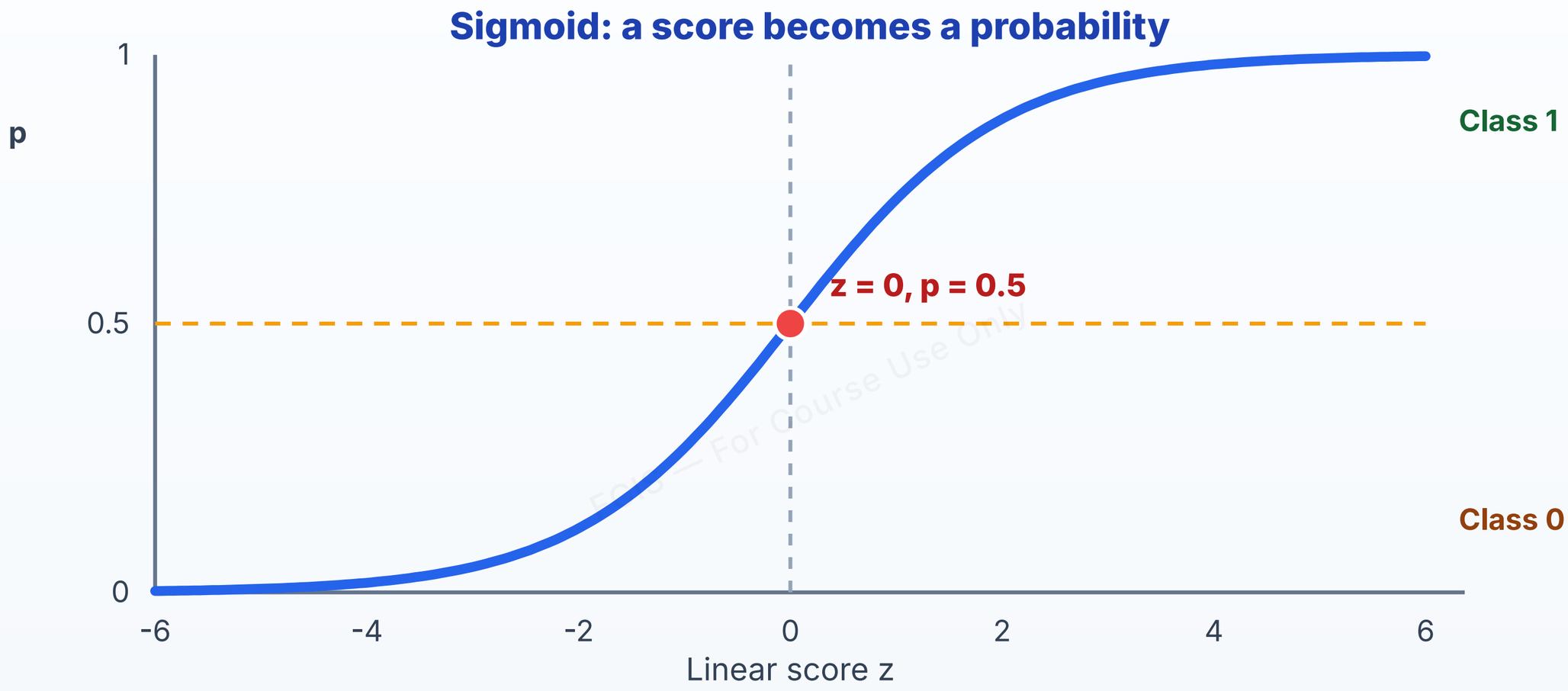
z	$\sigma(z)$
very negative	close to 0
0	0.5
very positive	close to 1

💡 The **sigmoid** maps every real-valued score into the interval $(0, 1)$.

Visualizing the Sigmoid

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A continuous probability, then a decision threshold



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💡 A zero score corresponds to probability 0.5. The curve saturates near 0 and 1; predicted probabilities should still be checked for **calibration** on held-out data.

Sigmoid Interpretation

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Probability and direction

The sign and magnitude of z both matter.

Score z	Probability p	Interpretation
$z = -4$	close to 0	model strongly favors class 0
$z = -1$	below 0.5	moderate evidence for class 0
$z = 0$	0.5	uncertain boundary
$z = 1$	above 0.5	moderate evidence for class 1
$z = 4$	close to 1	model strongly favors class 1

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Changing z from 0 to 2 changes the probability more than changing z from 8 to 10 because the **sigmoid** saturates. **A large model probability is not automatically a calibrated real-world probability.**

Worked Forward Pass

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One prediction

Suppose a classifier uses: $z = 0.8x_1 - 0.4x_2 + 0.2$

For:

$$x_1 = 2, \qquad x_2 = 1$$

the score is:

$$z = 0.8(2) - 0.4(1) + 0.2 = 1.4$$

The predicted probability is:

$$p = \sigma(1.4) = \frac{1}{1 + e^{-1.4}} \approx 0.802$$

With threshold 0.5: $\hat{y} = 1$

> The model predicts class 1 with probability about 80.2%.

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Odds and Log-Odds

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What logistic regression is linear in

If:

$$p = P(y = 1 \mid x)$$

then the odds are:

$$\frac{p}{1 - p}$$

The log-odds are:

$$\log\left(\frac{p}{1 - p}\right)$$

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Logistic regression assumes the **log-odds** are linear:

$$\log\left(\frac{p}{1 - p}\right) = w^\top x + b$$

💡 The probability is nonlinear in x, but the log-odds are linear in x.

Decision Rule

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Turning probability into a class

For threshold t:

$$\hat{y} = \begin{cases} 1, & p \geq t \\ 0, & p < t \end{cases}$$

The default threshold is often:

$$t = 0.5$$

But the best threshold depends on the application.

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Example

In fraud detection, missing fraud may be much more costly than temporarily flagging a legitimate transaction.

> 💡 Threshold selection is a decision problem, not only a modeling problem.

Part III

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Decision Boundaries

Logistic regression separates classes using a linear boundary in the feature space.

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Boundary at Threshold 0.5

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Where the model is uncertain

At threshold 0.5: $p = 0.5$

Since: $\sigma(0) = 0.5$,

the **decision boundary** occurs when:

$$w^\top x + b = 0$$

In two dimensions

If:

$$z = w_1 x_1 + w_2 x_2 + b$$

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then the boundary is the line:

$$w_1 x_1 + w_2 x_2 + b = 0$$

> Points on one side are predicted class 1; points on the other side are predicted class 0.

Changing the Threshold

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Same model, a different decision boundary

For logistic regression, $p = \sigma(z)$ and $z = w^\top x + b$. A threshold t predicts class 1 when $p \geq t$.

Because **sigmoid** is increasing, this is equivalent to:

$$w^\top x + b \geq \log\left(\frac{t}{1-t}\right)$$

Thus the boundary changes from $z = 0$ at $t = 0.5$ to $z \approx 1.099$ at $t = 0.75$.

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💡 Changing the **threshold** changes **which scores count as positive**, not the trained coefficients. We select the threshold using validation data and the costs of mistakes.

Worked Boundary Example

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Interpret a two-feature classifier

Suppose: $z = 2x_1 + x_2 - 6$

At threshold 0.5, the boundary is:

$$2x_1 + x_2 - 6 = 0$$

So:

$$x_2 = 6 - 2x_1$$

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Classification rule

If: $2x_1 + x_2 - 6 \geq 0$

then: $\hat{y} = 1$

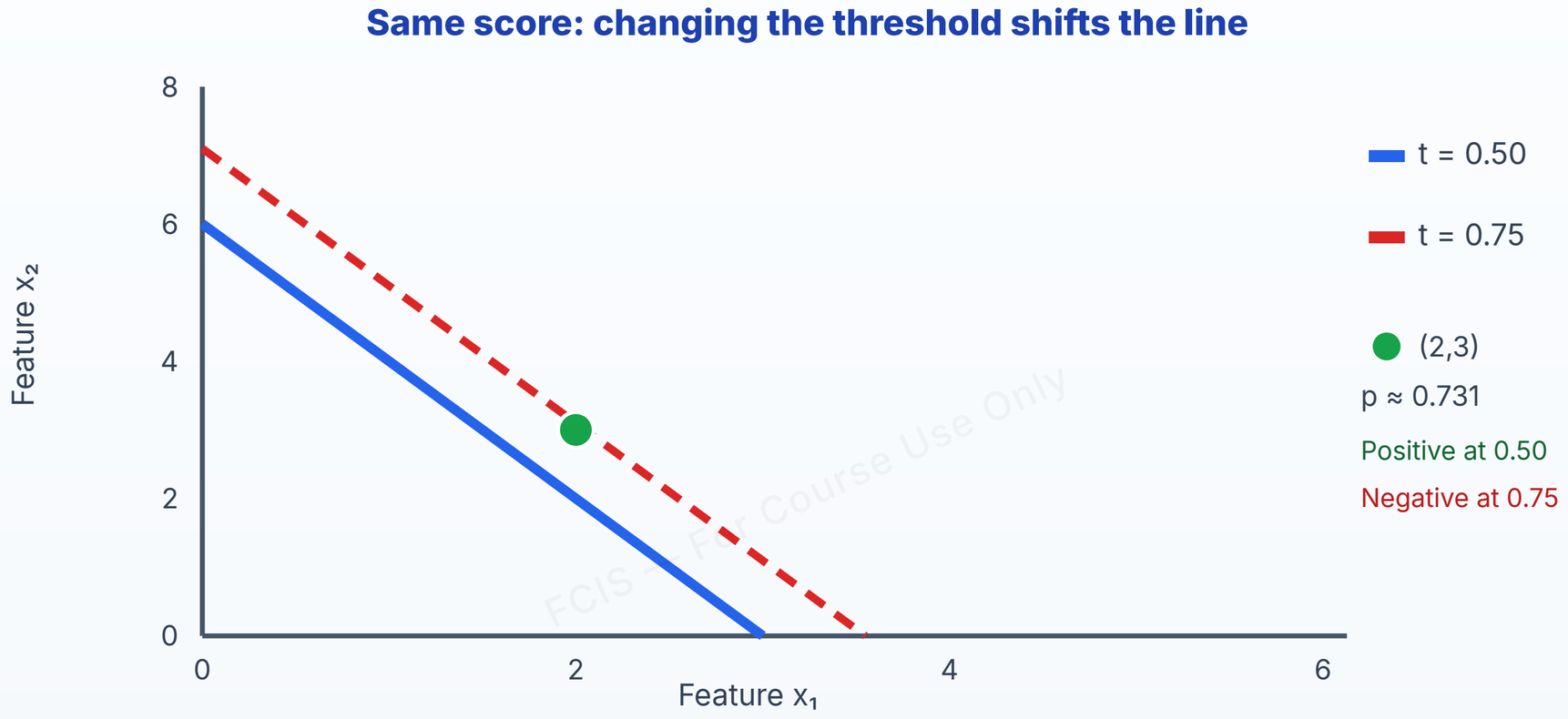
Otherwise: $\hat{y} = 0$

> For $x_1 = 2$ and $x_2 = 3$, $z = 1$, so $p = \sigma(1) \approx 0.731$ and the predicted class is 1.

Visualizing Threshold-Dependent Boundaries

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Changing a threshold changes decisions without retraining



For $z = 2x_1 + x_2 - 6$, the green example $(2, 3)$ has $z = 1$ and $p \approx 0.731$: class 1 at $t = 0.5$, class 0 at $t = 0.75$. The model weights and bias have not changed.

Effect of Coefficients

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Direction and strength

In:

$$z = w_1 x_1 + \cdots + w_p x_p + b$$

the sign of w_j affects the probability.

Coefficient	Effect when feature increases
$w_j > 0$	increases z, usually increases $P(y = 1 \mid x)$
$w_j < 0$	decreases z, usually decreases $P(y = 1 \mid x)$
$w_j = 0$	no direct effect in the model

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Caution

Coefficient size depends on feature scale, and comparisons are meaningful only with other features held fixed.

> 💡 Standardization makes coefficient magnitudes more comparable, but interpretation still needs domain knowledge.

Linear Separability

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When a straight boundary is not enough

Logistic regression creates a linear decision boundary in the transformed feature space.

It works well when classes are approximately separable by a line or hyperplane.

If the relationship is nonlinear

Possible responses:

- add interaction features;
- add polynomial features;
- use a different model such as decision trees or neural networks;
- transform the feature representation.

> Feature engineering can make logistic regression more flexible without changing the algorithm itself.

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Part IV

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Binary Cross-Entropy Loss

For classification, the loss should reward correct probabilities and strongly penalize confident wrong predictions.

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Why Not Squared Error?

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Classification needs a probability loss

Squared error can be used with probabilities, but it is not the standard choice for logistic regression.

We want a loss that:

- encourages high probability for the correct class;
- penalizes confident wrong predictions strongly;
- works naturally with the sigmoid model;
- leads to convenient gradients.

This leads to **binary cross-entropy**, also called log loss.

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> 💡 A classifier should not only predict labels; its probabilities should also support meaningful decisions. Log loss encourages better probability estimates, but calibration still needs to be checked.

Binary Cross-Entropy

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Loss for one example

For true label:

$$y \in \{0, 1\},$$

and predicted probability:

$$p = P(y = 1 \mid x)$$

binary cross-entropy is:

$$L(y, p) = -\left[y \log(p) + (1 - y) \log(1 - p)\right]$$

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Cases

If $y = 1$:

$$L = -\log(p)$$

If $y = 0$:

$$L = -\log(1 - p)$$

Interpreting Log Loss

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Confident wrong predictions are expensive

True label	Predicted probability for class 1	Loss
1	0.90	$-\log(0.90) \approx 0.105$
1	0.60	$-\log(0.60) \approx 0.511$
1	0.10	$-\log(0.10) \approx 2.303$
0	0.10	$-\log(0.90) \approx 0.105$
0	0.90	$-\log(0.10) \approx 2.303$

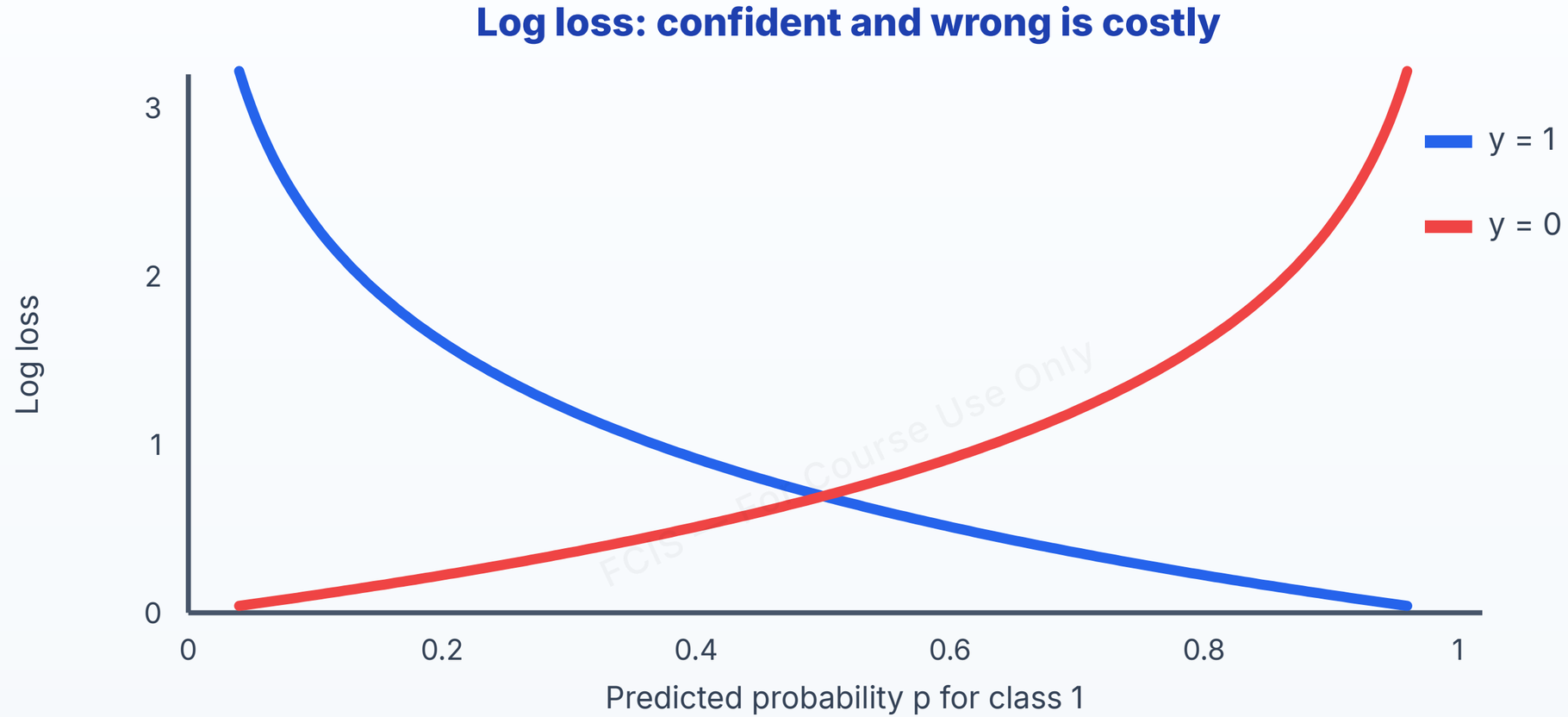
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Being wrong with high confidence is penalized much more than being uncertain.

Visualizing Binary Cross-Entropy

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Why confidently wrong predictions are heavily penalized



💡 For a truly positive case ($y = 1$), loss falls as p rises; for a truly negative case ($y = 0$), it rises as p rises. The vertical axis shows the **per-example loss**, before averaging across the dataset.

Dataset Loss

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Average over examples

For n training examples, the average **binary cross-entropy** is:

$$J(w, b) = -\frac{1}{n} \sum_{i=1}^{n} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

where:

$$p_i = \sigma(w^\top x_i + b)$$

Learning means choosing w and b to minimize $J(w, b)$.

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> 💡 This is the objective optimized during logistic-regression training.

Part V

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Learning Logistic Regression

Training changes the weights and bias so that predicted probabilities better match the observed labels.

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Forward Pass

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From input to probability to loss

For one example:

1. Compute score:

$$z = w^\top x + b$$

2. Compute probability:

$$p = \sigma(z)$$

3. Compute loss:

$$L(y, p) = -\left[y \log(p) + (1 - y) \log(1 - p)\right]$$

4. Use gradients to update the parameters.

> The same pattern will reappear later in neural networks: forward pass, loss, backward pass, update.

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Gradient for One Example

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Useful compact result

For **logistic regression** with binary cross-entropy, the derivative with respect to the score is:

$$\frac{\partial L}{\partial z} = p - y$$

Then:

$$\frac{\partial L}{\partial w_j} = (p - y)x_j$$

and:

$$\frac{\partial L}{\partial b} = p - y$$

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Interpretation

- if $p > y$, the model predicted too high;
- if $p < y$, the model predicted too low.

💡 The quantity $p - y$ is the probability error signal.

Gradient Descent Update

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One-example update

For learning rate α:

$$w_j \leftarrow w_j - \alpha(p - y)x_j$$

$$b \leftarrow b - \alpha(p - y)$$

In vector form:

$$w \leftarrow w - \alpha(p - y)x$$

$$b \leftarrow b - \alpha(p - y)$$

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The update pushes the model to increase probability for positive examples and decrease it for negative examples.

Worked Training Step

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Forward pass

Suppose:

$$x = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \qquad y = 1$$

and current parameters:

$$w = \begin{bmatrix} 0.5 \\ -0.25 \end{bmatrix} \qquad b = 0.1$$

Score:

$$z = w^\top x + b = 0.5(2) + (-0.25)(1) + 0.1 = 0.85$$

Probability:

$$p = \sigma(0.85) = \frac{1}{1 + e^{-0.85}} \approx 0.701$$

Loss:

$$L = -\log(0.701) \approx 0.355$$

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Worked Training Step

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Backward pass and update

The error signal is: $p - y = 0.701 - 1 = -0.299$

Gradients:

$$\frac{\partial L}{\partial w_1} = (-0.299)(2) = -0.598,$$

$$\frac{\partial L}{\partial w_2} = (-0.299)(1) = -0.299,$$

$$\frac{\partial L}{\partial b} = -0.299.$$

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With learning rate: $\alpha = 0.1$

updates are:

$$w_1 \leftarrow 0.5 - 0.1(-0.598) = 0.5598,$$

$$w_2 \leftarrow -0.25 - 0.1(-0.299) = -0.2201,$$

$$b \leftarrow 0.1 - 0.1(-0.299) = 0.1299.$$

Because the true class is 1 and p was too low, the update increases the score for this example.

After the Update

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Did the probability improve?

Using the updated parameters:

$$w = \begin{bmatrix} 0.5598 \\ -0.2201 \end{bmatrix}, \qquad b = 0.1299.$$

For the same input:

$$z = 0.5598(2) - 0.2201(1) + 0.1299 = 1.0294.$$

New probability:

$$p = \sigma(1.0294) \approx 0.737.$$

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Comparison

Before update: $p \approx 0.701$, so $L = -\log(p) \approx 0.356$.

After update: $p \approx 0.737$, so $L = -\log(p) \approx 0.305$.

The probability for the correct class increased and **the example's log loss decreased** after one training step. This is one step on one example; improvement on the whole dataset is not guaranteed after every individual step.

Batch Gradients

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Average error signal over the training set

For n examples:

$$p_i = \sigma(w^\top x_i + b)$$

The average gradient is:

$$\frac{\partial J}{\partial w_j} = \frac{1}{n} \sum_{i=1}^{n} (p_i - y_i) x_{ij}$$

and:

$$\frac{\partial J}{\partial b} = \frac{1}{n} \sum_{i=1}^{n} (p_i - y_i)$$

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Update

$$w_j \leftarrow w_j - \alpha \frac{\partial J}{\partial w_j}$$

Logistic Regression Training Algorithm

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Pseudocode

```
Initialize weights w and bias b

repeat for several epochs:
    for each training example or mini-batch:
        compute z = wᵀx + b
        compute p = sigmoid(z)
        compute gradients using p − y
        update w and b using gradient descent

return trained parameters
```

Training hyperparameters

- learning rate;
- number of epochs;
- regularization strength;

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Separate validation-time choices

- decision threshold;
- model and preprocessing choices, assessed without using the final test results.

> 💡 The decision threshold is **not** updated by **gradient descent**. Choose it using validation performance and application costs; fit preprocessing statistics on training data only.

Part VI

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Thresholds, Decisions, and Probability Quality

A probability prediction becomes useful only when connected to a **decision rule**.

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Thresholds

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The default is not always best

Using $t = 0.5$:

$$p \geq 0.5 \Rightarrow \hat{y} = 1.$$

But other thresholds are possible.

Lower threshold

- more examples predicted positive;
- cannot decrease recall on the same labeled dataset;
- may increase **false positives**.

Higher threshold

- fewer examples predicted positive;
- may improve precision, but this is not guaranteed;
- may increase **false negatives**.

> For disease screening, a lower threshold may be preferred to reduce missed cases.

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Overlapping Class Scores with Python

For Course

Why changing the threshold trades false alarms for missed cases

This **synthetic score example** shows overlap between the two true classes. The two thresholds are alternative policies applied to the **same** scores.



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💡 **Takeaway.** Moving the threshold right reduces the number flagged, but may miss more actual positives. A threshold changes the decision, not the fitted probabilities.

Threshold Example

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Fraud detection

Predicted probabilities:

Transaction	$P(\text{fraud})$
A	0.91
B	0.64
C	0.48
D	0.22
E	0.07

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If threshold is 0.5

Flag: $A,\ B$

If threshold is 0.2

Flag: $A,\ B,\ C,\ D$

💡 Lowering the threshold flags more transactions. Without the true labels, we cannot yet calculate how many extra flags are genuine fraud or **false positives**.

Confusion Matrix Preview

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Detailed metrics come next lecture

Binary decisions can be summarized as:

	Predicted 1	Predicted 0
Actual 1	True Positive	False Negative
Actual 0	False Positive	True Negative

Why this matters

Different errors may have different costs.

- False negative in fraud detection: fraud is missed.
- False positive in fraud detection: legitimate transaction is delayed.

> Lecture 5 will study metrics such as accuracy, **precision**, recall, F1, and cost-sensitive evaluation in detail.

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Probability Calibration

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Does 0.8 really mean 80%?

A classifier is well calibrated if among examples assigned **probability** about 0.8, approximately 80% are truly positive.

Why calibration matters

- medical risk estimation;
- loan default probability;
- fraud risk scoring;
- decision thresholds;
- resource prioritization.

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Caution

Logistic regression often gives useful probabilities, but **calibration** should still be checked when probabilities are used for decisions.

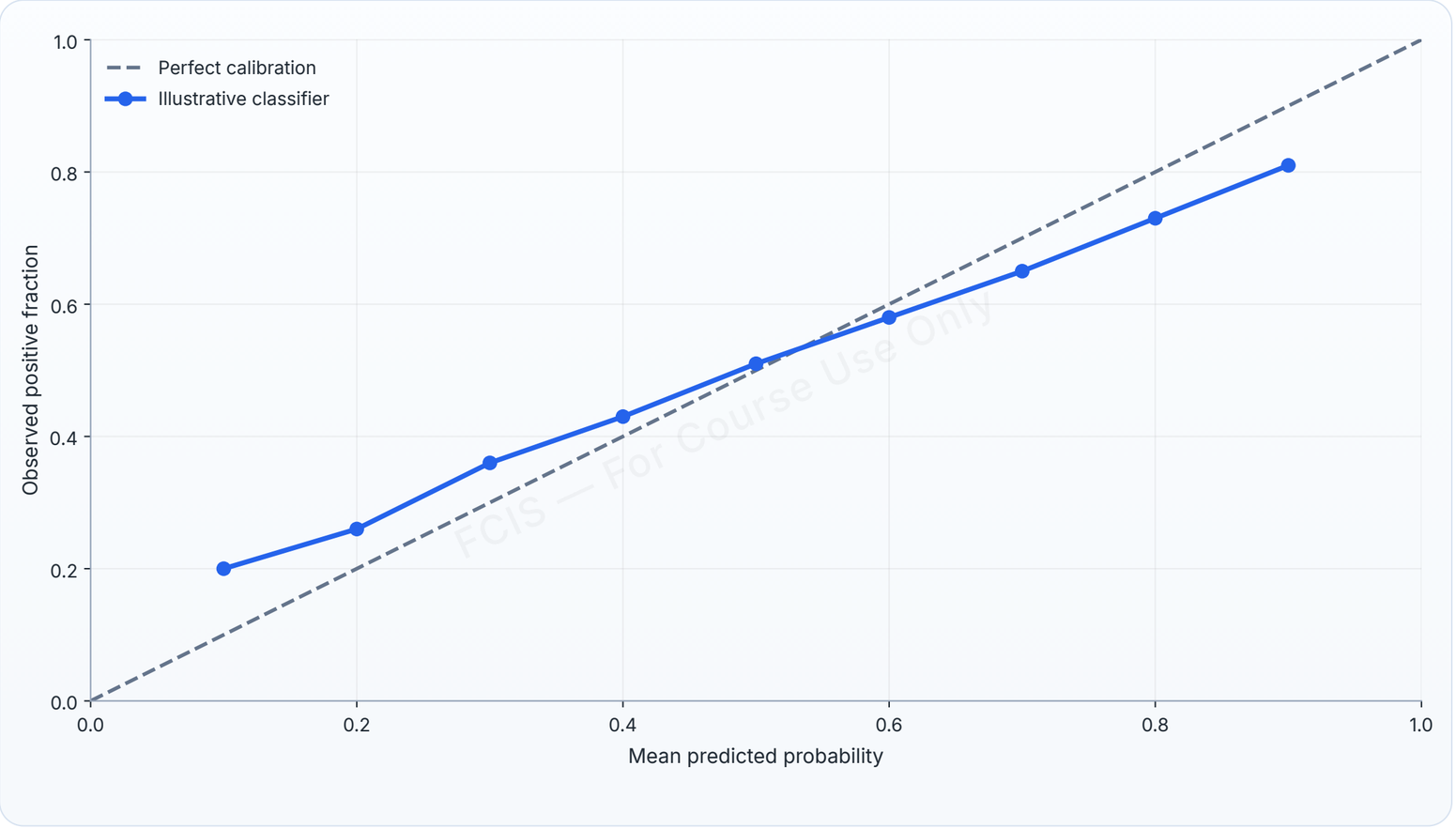
> 💡 Classification accuracy and probability quality are not exactly the same thing.

Probability Calibration with Python

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A score of 0.8 should correspond to roughly 80% positives

Each point represents a **hypothetical probability bin**. The diagonal is perfect **calibration**; deviations mean predicted probabilities do not match observed frequencies.



💡 **Takeaway.** An apparently confident classifier can be poorly calibrated. Check probability quality independently from classification accuracy.

Class Imbalance

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Accuracy can be misleading

Suppose only 1% of transactions are fraudulent.

A model that predicts every transaction as legitimate gets: $99\%\ \text{accuracy}$.

But it detects: $0\%\ \text{of fraud}$

Implication

For imbalanced classification, we often care about:

- recall;
- precision;
- false-negative cost;
- false-positive cost;
- threshold selection.

> High accuracy does not necessarily mean a useful classifier.

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Part VII

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Interpreting Logistic Regression

Logistic regression is often attractive because its coefficients and decision boundary can be interpreted.

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Coefficient Interpretation

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Direction of effect

In logistic regression, let $p = P(y = 1 \mid x)$:

$$\log\left(\frac{p}{1-p}\right) = w^\top x + b$$

- If $w_j > 0$, increasing x_j **increases** the predicted probability of class 1.
- If $w_j < 0$, increasing x_j **decreases** the predicted probability of class 1.

For models with multiple features, these interpretations assume the other features remain fixed.

Example: Predicting loan default

Suppose a bank predicts **loan default** (1 = default, 0 = no default) using one feature:

Debt-to-income ratio (DTI) = Monthly debt payments / Gross monthly income.

A **positive DTI coefficient** means a higher debt-to-income ratio is associated with a **higher predicted probability of default**.

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> 💡 **Takeaway.** Coefficient signs indicate the **direction of association**, not causation. Magnitudes depend on feature scaling.

Odds Ratio

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Exponentiating coefficients

For one feature x_j, increasing x_j by one unit changes the odds by a factor of:

$$e^{w_j}$$

Example

If:

$$w_j = 0.7$$

then:

$$e^{0.7} \approx 2.01$$

The **odds**, not the probability, are multiplied by about 2 for a one-unit increase in x_j, holding other features fixed.

> This interpretation depends strongly on the unit and scale of x_j.

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Interpreting Encoded Categories

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Coefficients depend on representation

Suppose `Learning Mode` is one-hot encoded:

Feature	Meaning
`mode_group`	1 if group learning
`mode_online`	1 if online learning

If `individual` is the reference category, coefficients compare each category to individual learning.

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Example

A positive coefficient for `mode_group` means group learning increases the log-odds of the **positive class** compared with individual learning, holding other variables fixed.

> 💡 Interpretation requires knowing which category was omitted as the reference.

Regularized Logistic Regression

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Preventing overly large coefficients

As in Lecture 3, **logistic regression** can be regularized to manage model complexity. This reuses the familiar idea while changing the **data-fit loss** from squared error to binary cross-entropy.

A common objective is:

$$J_{\text{reg}}(w, b) = J(w, b) + \lambda \sum_{j=1}^{p} w_j^2.$$

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Why regularize?

- reduce overfitting;
- handle many features;
- stabilize coefficients;
- improve generalization.

Text classification with thousands of word features often benefits from regularization.

Part VIII

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Preparing Features for Logistic Regression

The model is simple, but the representation must still be carefully prepared.

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Numerical Features

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Scaling and interpretation

Scaling is often useful because:

- **gradient descent** may converge more easily;
- regularization treats coefficients more fairly;
- coefficients become more comparable;
- numerical stability can improve.

Common choices

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$$z = \frac{x - \mu}{\sigma}$$

or:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

💡 Fit scaling parameters on the training set only, then transform validation and test data using those same parameters.

Categorical Features

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Encoding is required

Logistic regression needs numerical input.

Nominal categorical variables are usually represented using one-hot encoding.

Example:

City	Madinah	Riyadh	Jeddah
Madinah	1	0	0
Riyadh	0	1	0
Jeddah	0	0	1

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Caution

High-cardinality categories can create many features and overfitting risk.

Nonlinear Patterns

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Feature engineering can help

Logistic regression has a linear decision boundary in the feature space.

To model nonlinear behavior, we may add:

- polynomial features;
- interaction features;
- log transformations;
- domain-specific engineered variables.

Example interaction:

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$$x_{\text{interaction}} = x_1 x_2$$

A simple model with good features can sometimes outperform a complex model with poor representation.

Part IX

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Beyond Binary Classification

Many real classification tasks have more than two possible classes.

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From Binary to Multiclass

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More than two labels

Examples:

Problem	Classes
Digit recognition	0, 1, 2, ..., 9
Email routing	academic, finance, IT, administration
Medical diagnosis	disease A, disease B, disease C
Sentiment	negative, neutral, positive

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Two common strategies:

1. One-vs-rest classification
2. Softmax regression

💡 The central idea remains the same: compute scores, convert them into class decisions, and train using a suitable loss.

One-vs-Rest

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Train one classifier per class

For K classes, train K binary classifiers.

Each classifier asks:

> Is this example class k or not?

Prediction chooses the class with the highest score or probability.

Advantages

- simple extension of binary **logistic regression**;
- easy to understand;
- works with many binary classifiers.

Limitation

The separate probabilities may not naturally sum to 1.

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Softmax Regression

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Probability distribution across classes

For K classes, compute a score for each class:

$$z_k = w_k^\top x + b_k$$

Softmax converts scores into probabilities:

$$P(y = k \mid x) = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}$$

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The probabilities sum to 1:

$$\sum_{k=1}^{K} P(y = k \mid x) = 1$$

> Softmax regression is the multiclass generalization commonly used in neural-network output layers.

Part X

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Applied Mini-Case: Student Success

Let us connect the logistic-regression pipeline using a student-success classification problem.

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Mini-Case: Student Success Prediction

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Problem formulation

A university wants to predict whether a student is at risk of failing a course.

Features

Feature	Type	Preprocessing
Attendance percentage	numerical	scaling
Number of missed labs	discrete numerical	scaling or count
Previous GPA	numerical	scaling
Learning mode	categorical	one-hot encoding
LMS activity score	numerical	scaling

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Target

$$y = \begin{cases} 1, & \text{at risk} \\ 0, & \text{not at risk} \end{cases}$$

💡 The predicted probability should support advising decisions, not automatically punish the student.

Mini-Case: Model Output

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From probability to action

Suppose the model outputs:

Student	$P(\text{at risk})$	Threshold 0.5	Threshold 0.25
A	0.72	at risk	at risk
B	0.46	not at risk	at risk
C	0.28	not at risk	at risk
D	0.12	not at risk	not at risk

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Questions

1. Which threshold leads to more interventions?
2. Which threshold may reduce missed at-risk students?
3. Which threshold may increase unnecessary interventions?

> These are hypothetical predictions. Lowering the threshold increases interventions; whether it catches more truly at-risk students depends on the actual outcomes.

Mini-Case: Check the Decisions

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A small, illustrative validation example

Assume we later observe the **true at-risk labels** for these same four students. These labels are fictional and are used only to illustrate threshold reasoning.

Student	Model probability	True at-risk label	At $t = 0.5$	At $t = 0.25$
A	0.72	1	Positive	Positive
B	0.46	0	Negative	Positive
C	0.28	1	Negative	Positive
D	0.12	0	Negative	Negative

Measure (positive = at risk)	$t = 0.5$	$t = 0.25$
True positives / false negatives	1 / 1	2 / 0
False positives / true negatives	0 / 2	1 / 1
Recall	$1/2 = 50\%$	$2/2 = 100\%$
Precision	$1/1 = 100\%$	$2/3 \approx 67\%$

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💡 The lower threshold catches student C but also unnecessarily flags student B. **Four invented cases are too few to establish a real threshold's quality**; use representative **validation data** and consider intervention costs and benefits.

Mini-Case: Reasoning Questions

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What should students explain?

1. Why is this a classification problem rather than regression?
2. Why does the model output a probability before a class label?
3. Why might a threshold lower than 0.5 be reasonable?
4. Which features may create leakage if measured after the final exam?
5. Why should the model be evaluated on future or held-out students?
6. How can **false positives** and false negatives have different consequences?

> A useful classifier must be evaluated not only mathematically, but also in its decision context.

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Practice & Consolidation

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Logistic Regression: Activities and Common Mistakes

Logistic regression is mathematically simple, but its predictions and thresholds require careful interpretation.

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Activity: Compute a Probability

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Manual practice

Suppose:

$$z = 1.2x_1 - 0.8x_2 - 0.4$$

For:

$$x_1 = 1, \qquad x_2 = 2$$

compute:

1. the score z;
2. the probability $p = \sigma(z)$;
3. the predicted class using threshold 0.5.

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Manual practice

Suppose:

$$z = 1.2x_1 - 0.8x_2 - 0.4$$

For:

$$x_1 = 1, \qquad x_2 = 2$$

compute:

1. the score z;
2. the probability $p = \sigma(z)$;
3. the predicted class using threshold 0.5.

$z = 1.2(1) - 0.8(2) - 0.4 = -0.8.$
$p = \sigma(-0.8) \approx 0.310.$
At threshold 0.5, the predicted class is 0.

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Activity: Compute Log Loss

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One example

For each case, compute the **binary cross-entropy** loss.

Case	True label y	Predicted probability p	Loss
A	1	0.9	
B	1	0.2	
C	0	0.1	
D	0	0.8	

Questions

1. Which predictions are confident and correct?
2. Which predictions are confident and wrong?
3. Why is **log loss** useful for probability models?

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One example

For each case, compute the **binary cross-entropy** loss.

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Questions

1. Which predictions are confident and correct?
2. Which predictions are confident and wrong?
3. Why is **log loss** useful for probability models?

$L_A = -\log(0.9) \approx 0.105$; $L_B = -\log(0.2) \approx 1.609$; $L_C = -\log(0.9) \approx 0.105$; $L_D = -\log(0.2) \approx 1.609$. A/C have low loss; B/D have high loss. Confident wrong predictions are penalized strongly.

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Activity: Threshold Choice

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Medical screening

A screening model predicts probability of disease.

Patient	Probability
P1	0.83
P2	0.58
P3	0.41
P4	0.27
P5	0.09

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Questions

1. Which patients are positive at **threshold** 0.5?
2. Which patients are positive at threshold 0.25?
3. Which threshold is more conservative?
4. Which threshold may be safer if missing a disease is very costly?

Medical screening

A screening model predicts probability of disease.

Patient	Probability
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Questions

1. Which patients are positive at **threshold** 0.5?
2. Which patients are positive at threshold 0.25?
3. Which threshold is more conservative?
4. Which threshold may be safer if missing a disease is very costly?

> Threshold 0.25 flags more patients and may reduce false negatives, but it may increase **false positives**.

Common Mistakes

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Logistic-regression errors

Mistake	Why it is a problem
Treating probability as certainty	ignores uncertainty
Using accuracy only on imbalanced data	can hide missed positives
Forgetting feature scaling with regularization	coefficients and optimization may be affected
Interpreting coefficients without knowing encoding	categorical effects may be misunderstood
Choosing threshold using final test data repeatedly	test set becomes part of model selection
Assuming threshold 0.5 is always best	ignores application costs
Using future information as a feature	leakage
Calling logistic regression a regression task	despite the name, it is usually used for classification

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Mini-Quiz

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Check your understanding

1. What does **logistic regression** estimate?
2. Why is the sigmoid function needed?
3. What is the decision boundary at threshold 0.5?
4. Why is log loss preferred over MSE for logistic regression?
5. What is the gradient error signal for one example?
6. Why can a lower threshold increase recall?
7. Why can class imbalance make accuracy misleading?
8. What is the difference between one-vs-rest and softmax?

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Check your understanding

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Suggested answers: $P(y = 1 \mid x)$; sigmoid maps scores to probabilities; $w^\top x + b = 0$; log loss penalizes confident wrong probabilities; $p - y$; more positives are flagged; a majority-class classifier can get high accuracy; one-vs-rest trains binary classifiers while softmax models a multiclass probability distribution.

Lecture 4 Summary

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Key ideas to retain

Model and learning

- **Logistic regression** is used for classification.
- It computes a linear score:

$$z = w^\top x + b$$

- It converts the score to a probability:

$$p = \sigma(z)$$

- Binary cross-entropy is the standard loss.
- With sigmoid plus log loss, the gradient signal is:

$$p - y$$

Decisions and interpretation

- A threshold converts probability into a class label.
- The default threshold 0.5 is not always optimal.
- Coefficients affect the log-odds of the positive class.
- Scaling and encoding affect both training and interpretation.
- Accuracy alone may be misleading, especially with imbalanced data.
- Multiclass classification can use one-vs-rest or softmax.

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End-of-Lecture Check

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Five questions to test your understanding

1. How does **logistic regression** transform a linear score into a probability?
2. Why does binary cross-entropy penalize confident wrong predictions strongly?
3. What happens to the decision boundary if the threshold changes (without retraining)?
4. Why might a medical screening system use a threshold lower than 0.5?
5. Why should coefficients be interpreted carefully when features have different scales?

Preparation for next lecture: Review TP, TN, FP, FN, precision, recall, F1-score, validation sets, and the idea of choosing a model based on application costs.

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Suggested Reading

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Primary and supporting references

Primary reference

G. James, D. Witten, T. Hastie, and R. Tibshirani, *An Introduction to Statistical Learning with Applications in Python*, Springer, 2023.

- Review sections on classification, logistic regression, odds, probabilities, and classifier interpretation.

Supporting references

A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly, 2022.

- Review classification, logistic regression, decision thresholds, and practical Scikit-Learn workflows.

K. P. Murphy, *Probabilistic Machine Learning: An Introduction*, MIT Press, 2022.

- Use selected sections for a deeper probabilistic view of logistic models and classification losses.

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💡 When reading, focus on the connection between probability prediction, loss, gradient update, and decision threshold.